

Institute of Actuaries of India

Subject SA1 – Health and Care Insurance

March 2018 Examination

INDICATIVE SOLUTION

Solution 1:

i) Various considerations before launching the product

Product Needs for consumers

- Need to understand the requirements of this product from customers' point of view
- Need to do the market research and understand demand for this product before launching
- Need to study success stories of other insurers for similar type of products in the market
- Need to check whether some part of the cost of dental treatment is covered by the Government or the employer.
- Need to understand whether customer is able to use mobile application of the company and whether he is comfortable in purchasing a policy using mobile application.

Shareholders

- Company would like to launch a product which is profitable to shareholders and hence profitability measures of this product should be assessed.
- Assessing the level of cost of capital using WACC or other appropriate approach
- As company has aggressive business plans, does company have adequate level of capital to support the business at a largescale?
- Shareholders would like to provide the level of capital support for this product which should be optimal as shareholders may not favourable more consumption of capital compared to other products.
- Shareholders would also like to assess the returns on capital deployed for this product.
- Reinsurance arrangement may help to reduce the capital requirements and also reduce the claim volatility. Type of reinsurance arrangement-quota share, surplus, stop loss etc can be explored within the regulatory framework.

Product Distribution and Service

- As this product is intended to be distributed through mobile application, we need to study the regulatory requirements for such innovation product.
- Whether company will sell this product directly through mobile application or whether company is considering distributing it through other distribution channels and what would be commission/ remuneration structure for these alternate distribution models.
- Whether the mobile application is user friendly and easy to issue the policy?
- Online videos and product feature materials need to be developed for customer education which can be accessed through mobile application.
- How the ongoing servicing of policy including claim management will be handled by the company? Whether there will be 24 x 7 support centres available to service the policyholders?

Product features

- Whether product is covering comprehensively or it is restrictive dental cover?

- Whether definitions of various dental diseases and treatments are mentioned unambiguously in the policy document?
- What about waiting period, deferred period, pre-existing conditions pertaining to tooth as part of exclusion clauses, coinsurance, limits, no claim discounts etc
- Underwriting philosophy, claims management-in house or using TPAs, investment philosophy, risk management framework , regulatory constraints, tax benefits by paying premiums/receiving benefits
- Whether this product is complimenting the existing insurance cover provided by the government or employer.
- Free dental check-ups for policyholders thereby cost incurred by the company for such check ups versus benefits of not claiming for insurance by policyholders
- Whether the product has unrestricted renewability clause and portability arrangements with other insurers and whether administrative system is capable to handle such aspects?

General points

- Administration system-System development, sales, underwriting, claims, customer service, MIS
- Estimation of expenses-launch, maintenance and claims expenses
- Expenses pertaining to development and testing of mobile application and also maintenance expenses of mobile application.
- Whether company has sufficient data to price the product or company is considering industry data or data from reinsurers adjusted for the local conditions.
- Integration of customer data with the core IT system of the company
- Whether the premium is fixed for certain number of years or renewable every year?
- Adherence to Treating Customers Fairly
- Developments of actuarial models particularly for pricing and valuation purpose
- Project volume and profitability assessment considering various scenarios and understanding impact of current and expected future level of solvency pertaining to this product

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ii) Identify the risks around this product and how do you mitigate the risks.

Risk Identification

- As policy will be issued online, there could be a big underwriting risk as company may be exposed to adverse selection and non disclosure by prospects.
- As this is an indemnity based product, there is a risk that the frequency and severity of claim payouts could be much higher than anticipated.
- Inaccurate, incomplete and inadequate policy data may be captured.
- Inaccurate estimation of actuarial assumptions due limited availability of morbidity data and medical inflation data. There is also model risk, parameter risk and random fluctuation risk.

- Risks from writing too little business leading to new business strain and increase of per policy expenses or writing too much business may lead to consumption of capital. There is also a risk from competitors leading to shrinking of market share for this product.
- Legal & regulatory risk as new regulations may restrict to sell this type of product..
- Counterparty risks under reinsurance arrangement, Operation risk, morbidity risk.
- Counterparties in the provision of medical services, in that they can have influence over claims costs and the client relationship.
- Reputational risks, internal audit failure, fraud, policy wordings risk, non disclosure, advances in dental science leading to increasing in dental claims.
- High risk of policy non renewal, expense overruns, product design failure.

Risk Mitigation

- Product related risks can be mitigated by introducing waiting period, deferred period, coinsurance, benefit limits, exclusion of pre existing conditions pertaining to tooth, unambiguous terms and conditions, introduction of no claim bonus, free dental check up on regular basis.
- Negotiating the cost of treatment for various dental related diseases with the dental clinics.
- From risk diversification point of view, selling this product across all parts of the country
- Surprise visits by the company officials to various empaneled dental clinics to check whether the claims are genuine or not.
- Mitigating anti selection risk by having robust back- end /claim underwriting
- Adequate risk classification within the underwriting process will help to ensure that all risks are rated fairly.
- Adding additional margins in the premiums to compensate for model, parameter and random fluctuation risks.
- Regularly reviewing the definitions of the medical diseases pertaining to tooth to avoid windfall payments.
- If capital constraint is there then the product can be sold for limited time period to avoid any further initial strain.
- Have a dedicated risk department to identify, monitor and mitigate any risks arising due to this product.
- Have reinsurance arrangements leading to lower capital requirement, risk sharing with reinsurer, smoothing of profits, technical and underwriting support from reinsurer.
- Badging arrangement.

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iii) After doing the experience analysis, you have found that the overall claim experience across the portfolio of this business has not been favourable. However, for age band 20-40 of male policyholders in rural locations claim, experience has been very favourable. Your marketing director was impressed with this statistics and has proposed that, instead of revising the premium rates across portfolio, premium rates for each individual should be derived automatically using artificial intelligence system thereby policyholder with healthy tooth will be paying lower premiums compared to other policyholders. What consideration will you take before adopting the new technology?

- **Considerations before adopting a new technology**

- Increase in premiums based on unfavourable experience across portfolio may not be acceptable to all policyholders. If the premiums are increased based on unfavourable portfolio experience, then male policyholders between ages 20-40 in rural areas may not continue to renew the policy and may go elsewhere for seeking similar type of insurance cover. This may further worsen the claim experience of remaining policyholders in the portfolio.
- Whether the period of investigation is long enough to have credible experience or some events during the investigation period might have caused unfavourable claim experience such as use of inferior quality of toothpastes by urban policyholders, inflated claim amounts for some policyholders.
- Premiums are generally reviewed across all policies based on portfolio experience. Also it needs to be checked whether the data pertaining to rural male policyholders aged 20-40 is credible enough to reduce the premiums for this sub group.
- It is important to see whether the claim frequency or severity of claim or both are leading to unfavourable experience to entire portfolio and favourable experience to a sub group of male rural population with age band 20-40.
- It is important to see whether there is any difference in underwriting and claims management between urban and rural policyholders.
- It is good to adopt the technology like artificial intelligence. However, to apply such technology, we need to record data carefully and accurately at individual policy level such as age, gender, location, occupation, type of food intake, number of times brushing the teeth, frequency of visiting the dentist for regular check-ups, sum assured, number of claims made previously etc.
- Insurance is a pooling of risk concept and hence using new technology of artificial intelligence, all policyholders with good track record will be benefitted in terms of lower premiums but others may not be able to get an insurance cover at reasonable level of premiums.
- New technology may reduce underwriters' burden to quote the appropriate premiums automatically for the risk underwritten. However, for special and complex cases, underwriters can take an appropriate decision based on their experience.
- As claim experience of individual policyholder changes over the period of time along with change in other attributes then it will be difficult for the insurance company to explain the new set of premiums for every renewal based on complex and intelligent calculations performed automatically by the system.
- So, while setting the premiums at every renewal, blended approach of modern intelligence system along with traditional techniques could be thought to be adopted.
- Need to see whether there are any regulatory restrictions or encouragement for using the artificial intelligence system to derive the premiums.
- Need to see whether competitors are adopting a new technology where premium will be based on individual policyholders' experience.
- Whether this suggested new technology can be used by inserting tiny device in the each policyholder's tooth and the condition of tooth will be monitored closely on

continuous basis and appropriate suggestions to protect the tooth will be sent automatically to policyholder. This may help to reduce the actual cost of claim. However, company needs to see the cost of purchasing this sophisticated device for each of the policyholders. On the other hand, not all policyholder's will be comfortable in fitting this device and it is too much administrative hassle to the company.

- Need to see the data protection and privacy laws of individual data collected by the insurance company.

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iv)

One toothpaste company has approached your company to market your dental insurance product and discount vouchers are offered with the purchase of toothpaste. Customer can buy insurance at discounted premium through using discount voucher. Discuss this proposal.

Need to understand why the toothpaste company is interested to offer discount voucher to its customers.

Need to see whether other insurance companies are offering similar types of discounts on such types of consumable goods like toothpaste, toothbrush, mouthwash liquid etc.

Assess whether the Toothpaste Company will receive any compensation from the insurer for the marketing activity of this product.

Whether the compensation payment to Toothpaste Company is as per the regulatory boundaries.

Morbidity risk pertaining to tooth of customers may differ if the target market in terms of socio-economic group or geographical region differs compared to what it was assumed in the pricing.

There is no data available for the conditions of tooth of the customers who purchase the toothpaste. Hence additional margins are required while setting reserves for this portfolio of policyholders.

Discount vouchers may increase the sell of the product but it may lead to higher capital requirement and higher marketing expenses to write new business.

On the contrary, this marketing activity may be a total failure leading to increase in expenses per policy for the company and may lead to less profit making or loss making proposition.

Sales staff must be trained to explain the features of the product to customers who are intending to buy the toothpaste.

It is generally expected that those customers who are purchasing the toothpaste are expecting to keep their tooth in a healthy condition. Hence, it is appropriate to offer discount on the premiums for the insurance cover.

However, if the quality of toothpaste is substandard or if customers do not do the proper brushing or gargling on regular basis even after using the toothpaste then it may lead to various tooth related diseases and hence it may not make sense to offer discounts on the premiums to customers of the toothpaste company.

There is a reputational risk to the insurer if the quality of toothpaste is not good or if the market share of the toothpaste company reduces leading to reduce sell of the dental insurance product.

Also, not all the customers may utilize the discount vouchers due to lack of awareness of this product and lack of motivation to buy this product.

After getting discount vouchers from the toothpaste company, customers will have to go to mobile application of the insurance company.. So, those customers who do not like the brand or quality of this toothpaste company may not purchase the toothpaste and will not get the discounts on premiums compared to others who have purchased the toothpaste of the same company and utilized the voucher to get premium discounts.

Additional provision in the mobile application needs to be developed for those having discount voucher which may lead to additional cost to the company.

Need to understand whether company is achieving the target profitability for this segment of customers and overall profitability under this product.

Reinsurer may not be comfortable to offer reinsurance cover from this type of customers' profile and may quote higher reinsurance premiums.

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[40 marks]

Solution 2:

i) Cause of reduction in new business sales

The probable reasons for reduction in new business can be broadly divided into the following two categories:

- Reduction in volume of all individual PMI business sold
- Reduction in insurer's market share

Reduction in overall individual PMI market

1. Overall number of PMI purchasers reduced
2. Market premiums are perceived to be too high compared to the benefits being offered.
3. Reduced customer need
4. General improvement in welfare benefits
5. Economic uncertainty or job insecurity
6. Possible changes in tax benefits
7. Government initiatives in providing health care / insurance benefit which led to people opting out of their individual policies

Reduction in market share

1. Premium too high relative to competition or they may have reduced their premiums
2. Benefits are not competitive
3. Commission levels are uncompetitive

4. Arduous process of underwriting / policy issuance acting as a deterrent or resulting in sales leakage
5. Poor company reputation
6. Perception of poor customer service
7. Reduction in credit rating
8. Perception of strict claims philosophy
9. Poor relationship between intermediaries and customers.

[3]

ii) Following factors influence the demand of health insurance:

1. Level of competition and range of products available in market
2. Pricing levels and perceived value of money
3. The state of economy
4. The willingness and ability of intermediaries to sell insurance products
5. The appropriateness of the distribution methods to the products and customers
6. Public's attitude to risk
7. Levels of awareness of health issues and health insurance products driven by the media / advertising campaigns
8. Quality of customer services provided by insurers
9. Ability of products to meet customer needs
10. Employer's attitude to offering healthcare to employees
11. Levels of healthcare provision / benefits provided by state

[3]

iii) Impact of implementation of universal health program on private health insurance market

- Once the universal health care program is implemented, it is likely to increase the demand of health care in the country. This would be mainly because of increased awareness. As the government is proposing to cover only half the population, the program will be means tested which means only a certain section of the society will get the benefit. The rest of the population who may not be eligible and can afford to pay will likely to be attracted in buying health insurance.
- For the population that is covered under the program, the private health insurers may offer new products e.g. top up policies. as the amount of coverage may be limited, top-up policies will be beneficial to the insured.
- As a large number of people will seek treatment, there may be significant delays in getting treatment. This will be specially crucial for secondary and tertiary care where supply may be restricted. Hence, private insurance companies may offer a product that will provide treatment cost for those illnesses that have a long waiting period.
- In essence, the successful implementation of the program would lead to an increase in general awareness. That is likely to increase the demand of private health insurance by those who can afford or those excluded by the program.

- Given the likely increase in the demand of health care, the supply of health care too will increase. This is because the aim of the universal program will be to provide health cover to underprivileged sections of the society who live mostly in the remote / rural locations where health care is provided mostly by government agencies. These agencies may not be fully capable to handling additional demand and hence private health care providers will also create capacity in those areas. Once the supply side capacity is created, it will improve the access to health care for the general population. This is likely to increase the incidence of health insurance claims in general.
- Another aspect will be the underwriting restrictions as well benefit restrictions that may need to be revised. The universal health care program is likely to have very few exclusions or waiting period. It may also cover out-patient benefits as well as those who are currently uninsurable (HIV patients etc). The health insurers will be expected to provide similar benefits in their products. This may impact the product design as well as the premiums.
- Further, as the supply side (specially secondary and tertiary care) increase that will increase access, the frequency as well as cost of treatment is likely to increase. Hence, the overall health care cost is likely to rise. Also, access to tertiary care will increase the average claims cost under the health insurance as people will be able to access these facilities easily. In the short term, the demand may be higher than the supply which can push up the cost of medical care. In the medium to long term, the supply of the health will increase that will increase availability of health delivery mechanism and may reduce the overall cost of health care per capita due to economies of scale.
- Further, the government may also introduce treatment protocols as well as cost of certain treatment procedures and medical equipment (e.g. stents etc) in order to contain the costs.
- The government may also control introduction of new / specialized treatments if they are very expensive which could restrict medical advancements to a certain extent.
- If the program offered by government covers primary care, there may be pressure on insurers to offer similar benefit coverage to its customers.
- Overall the impact on private health insurance premiums will be limited in the immediate term but with change in benefit structure and relaxed underwriting, the premiums may increase in medium to long term.

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iv) Factors to be considered before changing policy conditions of an existing PMI policy

Following factors are to be considered before changing policy conditions for an existing PMI policy:

1. Legality
2. Market Perception
3. Projected claims
4. PRE / TCF

5. Administration
6. Reinsurer's consent

Legality

1. It may be difficult in worsening the terms of the contract
2. Guaranteed renewability- the health insurance regulations provide the customers right to renew the policy. So the policyholder may expect to have the same coverage on renewal.
3. Exact legal implications will depend on wordings of the revised policy conditions.
4. Regulator's view / approach regarding changing of policy conditions. The regulator may not allow the insurance company to change the policy condition if it is against the policyholder's interest.

Market Perception

1. Need care that approach is not out-of-line.
2. May be a strong competitive disadvantage if other insurers offer similar policy conditions.
3. Strong reputational risk specially if there are legal complaints against the insurer which gather media attention.

Projected Claims

1. Need to consider the impact of the change in overall claims. The strengthening of the policy wordings is likely to restrict those claims that are currently getting paid. The overall claim outgo should reduce as a result. More importantly it should also reduce the potential disputes as only genuine claims will get paid.
2. It has to be clearly established that the claims would reduce once the policy conditions are changed. The insurer may also need to decide whether to pass on the benefit to the customer.
3. It may also result in selective lapses. Hence, the impact of claims and resultant premium has to be carefully assessed

PRE / TCF

1. Difficult to gauge for short-term policies
2. Depends on sales message at point-of-sale. Health insurance policies are yearly renewable which means that the terms can change on renewal. However, if the sales message doesn't clearly specify it then it creates a reasonable expectation amongst the policy holder.

Administration

1. Would need to be able to run two claims processes in parallel until the existing policies are renewed. The problem will be exacerbated if the insurer issues two or three year's policy.
2. Likely to result in practical difficulties

3. Will add to cost of processing claims. The claim assessors have to spend additional time in claims assessment.
4. May lead to contamination of old and new processes

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v) Factors that influence that approach to underwriting health products

By underwriting, the insurer attempts to classify the applicant into homogenous risk groups so that an appropriate price can be placed on each of the groups commensurate with the risks undertaken.

Following factors influence the approach to underwrite health products:

1. Range and size of benefits – the Sum insured levels as well as the benefits covered may influence the underwriting approach. For e.g the underwriting approach would differ if the underlying product offers only a flat sum insured of INR 1 Lakh vs if it offers a range of SI and the maximum sum insured is INR 1 crore.
2. Type of product – the underwriting approach will vary depending on the type of product. E.g hospital cash, surgical cash, indemnity product or critical illness
3. Method of distribution- The insurer may want to have differentiated underwriting approach depending on the distribution channel. This is because the socio-economic profiles of the customers vary depending on the distribution channel. E.g. the customers buying through online channel are likely to be much more aware than customers buying through agency channel. Hence the insurer may want to underwrite them differently.
4. Target market – the underlying product may be designed for a specific target segment of the population and hence underwriting approach can be set as per the perceived risk of the target underlying population .
5. Reinsurer's view – the reinsurer may also recommend the underwriting approach that is appropriate for the product and supported by the risk premiums
6. Approach taken by other insurer – if a similar product is already available in the market then the underwriting approach has to be designed keeping in mind the existing underwriting practices.
7. Level of competition – The underwriting approach will be influenced by the level of competition.
8. Regulatory constraints – the regulator may also influence the underwriting approach. E.g. genetic testing etc.
9. Cost vs benefit of underwriting – it is very important to establish the likely benefit of underwriting compared to the cost incurred.

[8]

vi) Following are the factors that may influence the propensity to claim by an individual

1. Current state of health – an individual who is healthy in the current state is likely to claim only either for infectitious diseases or due to accident.

2. Personal medical history – the medical history of the individual especially past incidences of chronic diseases will influence the claim behavior. hi
3. Family history – some of the diseases / conditions do have a strong correlation with family history (cancer, diabetes etc)
4. Financial circumstances - The financial circumstances of the individual may influence the propensity of claim. The tendency to claim will be high if the individual is under financial stress. Moreover, his /her ability to spend on preventive health care also depends on the financial circumstances.
5. Demographic details (age, sex, smoking habits) - older individuals are more likely to make a claim than younger individuals. Similarly, smokers are likely to get sick earlier than non-smokers.
6. Leisure / past time activities – individuals undertaking risky hobbies (mountaineering, scuba diving, racing etc) are more likely to claim than those who pursue painting or music as a hobby.
7. Area of residence / access to healthcare facilities - individual living in urban areas have better access to health facilities and are more likely to claim as compared to people living in rural areas who will have to travel to cities to get the treatment. Further individual living in a hygienic and clean environment are less likely to claim than otherwise.
8. Occupation – occupation has a strong association with propensity to claim. Individuals working in risky occupation like (mining, diving etc) are more likely to claim than people with desk jobs.

[5]

vii) Six actions that can be taken to deal with increased risk

1. Can be offered higher (loaded) premiums or reduction in benefit
2. Can be offered a reassessment following a deferral period (if the period of higher risk is deemed temporary)
3. Can be declined
4. Can be offered a different type of policy
5. Can be offered to a reinsurer facultatively with zero retention
6. Can have certain perils excluded

[3]

viii) Proposal evaluation

Impact on Sales

The proposal of offering relaxed underwriting to people over the age of 50 may help in company in boosting sales.

A quick analysis of the proposals of lives over age 50 may be undertaken to understand the proportion of policies that got either rated up, declined , postponed or were accepted as standard risk. The lives that were either declined or didn't accept the revised offer and hence cancelled the proposal will now be able to get covered as a result of relaxed underwriting.

Further, healthier lives will be attracted to come on board as they wouldn't need to undergo medicals and bear the cost of it as well.

If other insurers continue to have stringent underwriting requirements then by offering a relaxed underwriting, company can attract higher volumes of business.

Having said if there is no or limited initial underwriting, the underwriting will be mainly done at the time of claims. This could lead to disputes at the time of claim and may lead to litigation or bad reputation for the insurer.

To mitigate, the insurer can use tele-underwriting or pre-issuance verification call where the caller would inform about the benefits offered under the product as well as terms associated with it. If the called is medically trained then he/she can also seek more information about the health status of the individual and assess the risk.

Another way to mitigate anti-selection will be to use longer moratorium period.

Portfolio mix

If the insurer is the first to offer relaxed underwriting without change in premiums then it is likely to attract a majority of people over the age of 50. Hence, the average age of the portfolio is expected to increase.

Impact on Profitability

The impact of overall profitability will depend on the ultimate claims experience of lives that come on board due to relaxed underwriting. It is well understood that people over the age of 50 will have a heavier claims experience compared to younger ages but how much of it can be attributed to relaxed underwriting will have to be established. Also the length of selection effect will need to be analysed.

Without any medical underwriting, the claims are likely to start earlier than assumed in original pricing. Further, if a majority of the lives with substandard health are in the portfolio then the claims experience is likely to get worsened quickly. The problem can get further exacerbated given the lifelong renewability clause. Given the uncertainty involved, the insurer may opt for reinsurance that will help mitigate some of the downside risk.

Given the expectations of an older age distribution, if the underlying premiums were calculated assuming a younger age distribution then the premiums at the portfolio levels are likely to be insufficient. To mitigate, the insurer should undertake sensitivity analysis to assess the impact of different age distribution on the overall profitability.

On the other hand, if healthier lives too, as anticipated, join the portfolio then the overall experience is likely to be lighter than assumed and may be profitable.

Increased volumes will help the company to absorb the expenses and at the same time, the investment income (though marginal) may help the company to make profits even if the experience were to remain same.

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ix) Change in premium shape

Reduced premium for the older ages will certainly attract people of those ages, if it is the only insurer that is offering lower premiums at those ages. Given that most of these individuals are likely to be the decisions makers, it is possible that the sales for family floaters may also increase. This is likely to increase the average premiums as well as the sum insured. So, lower premiums along with limited underwriting may increase the overall sales and premiums.

The insurer should have the first movers' advantage till the other insurers follow suit.

Changing the premium shape would mean that the extent of cross subsidy between the ages will increase. This is likely to create portfolio mix risk. Premiums for younger lives are likely to be more expensive than the market that will make it less attractive and may reduce the sales at those ages. Further, those renewing the policies may also move to other insurers who are offering lower premiums. This lapses are likely to be anti-selective that will have poor impact on the claims experience.

In addition, existing policyholders who will reach the age of 50 will pay the same lower premiums as applicable to new policyholders. Since they have already been in the portfolio, the expected claims experience is likely to be heavier than the premiums paid by them. So if the proportion of new policyholders are lower than projected then the overall premiums may be insufficient and it could be loss making.

[6]

[60 Marks]
